A major project report on

SKILL PILOT AI: LLM POWERED CAREER COACH

Submitted in partial fulfillment of the requirements of the degree of

BACHELOR OF ENGINEERING

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CERTIFICATE

This is to certify that the project entitled **"Skill Pilot AI: LLM Powered Career Coach"** is a bonafide work of **Mithil Bhosale (22106134), Tejas Joshi (22106018), Saif Khan (22106072), Bhushan Khopkar (22106099)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning).**

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Project Report Approval for B.E.

This project report entitled ***"Skill Pilot AI: LLM Powered Career Coach"*** by **Mithil Bhosale (22106134), Tejas Joshi (22106018), Saif Khan (22106072), and Bhushan Khopkar (22106099)** is approved for the degree of ***Bachelor of Engineering*** in ***Computer Science & Engineering (Artificial Intelligence & Machine Learning), 2025-26***.

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Declaration

We declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honestly and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Abstract

The job market today changes very quickly. New technologies and different ways of working create a need for people to keep learning new skills throughout their lives. However, it can be difficult to get good career advice. Personal coaches are expensive, and online information can be overwhelming and not specific to one person's situation. This paper presents Skill Pilot AI, a concept for a new kind of career helper. It uses the power of advanced AI, specifically Large Language Models (LLMs), to act as a personal guide for career development.

Skill Pilot AI is designed to be a smart and easy-to-use tool for anyone looking to manage their career path. The system works by first talking to the user to understand their current skills, work experience, and future goals. It can look at a resume or just have a conversation to learn about a person's strengths and interests. Then, it connects this personal information with real-time data about the job market. This includes looking at which skills are in demand, what new jobs are appearing, and what companies are looking for.

Using all this information, the platform offers clear and useful advice. It can suggest possible career paths a person might not have considered. Most importantly, it identifies the specific gaps between a user's current skills and the skills needed for their desired job. For each skill gap, Skill Pilot AI creates a custom learning plan. It recommends online courses, articles, small projects, and other resources that are right for the user's level and needs.

In summary, Skill Pilot AI uses modern artificial intelligence to offer everyone access to personal career guidance, helping them navigate the modern working world successfully.

CONTENTS

1. Introduction ……..1

2. Literature Survey ……..4

3. Limitations of the Existing System ……...7

4. Problem Statement, Objective & Scope ……9

5. Proposed System ……12

6. Experimental Setup ……15

7. Results and Discussion ……...18

8. Conclusion ……20

9. Future Scope ……...21

References ……...22

Appendix

Publication

LIST OF FIGURES

6.1.1 System Design ……..……12

6.2.1 Login Page …..……..……13

6.2.2 Project Recommendation ……..……...13

6.2.3 Dashboard ……..……..…14

6.2.4 Gamefied Upskilling ……..……..14

ABBREVATION

AI	*Artificial Intelligence*
ML	*Machine Learning*

Chapter 1

Introduction

1. Introduction

Choosing a career path is one of the most important and challenging decisions a person makes in their life. In the past, a career was often a straight line: you learned a skill, got a job in a company, and worked there for many years, slowly moving up. Today, the world of work is completely different. It is dynamic, fast-paced, and constantly changing. New technologies like artificial intelligence and automation are creating new jobs while making others obsolete. This means that the skills needed for success today might not be enough tomorrow.

Because of this rapid change, people face more uncertainty and pressure than ever before. Students leaving school are unsure which fields will still be strong in five years. Professionals in the middle of their careers often feel stuck or worry that their skills are becoming outdated. The question "What should I do with my career?" is a common and difficult one to answer. This is where the role of a professional career coach becomes incredibly valuable.

A good career coach does much more than just help someone write a resume. They act as a guide, a supporter, and a strategist. Their important role involves several key activities:

- **Self-Assessment:** They help individuals understand themselves better. This means identifying their unique strengths, natural talents, personal values, and what truly interests them. A career that aligns with who you are is much more likely to be satisfying and successful.
- **Exploring Options:** The world has thousands of possible jobs and career paths. A coach helps explore these possibilities, many of which a person might not even know exist. They provide information about different industries, job roles, and what a typical day in those jobs looks like.
- **Goal Setting and Planning:** Once a direction is chosen, a coach helps break down the big, scary goal of "getting a new career" into small, manageable steps. They help create a clear roadmap, setting short-term and long-term objectives.
- **Skill Development Guidance:** They identify the gaps between a person's current skills and the skills required for their desired job. They then recommend specific courses, training programs, or experiences to bridge those gaps.

- **Support and Motivation:** Changing careers or starting a new one can be emotionally difficult. A coach provides encouragement, helps build confidence, to keep the person on track.

However, there is a big problem: professional career coaching is a service that is often expensive and not accessible to everyone. Most people cannot afford to hire a personal coach for regular sessions. This creates a situation where the people who need guidance the most cannot get it. This is the challenge that Skill Pilot AI aims to solve.

Challenges of Career Selection:

To understand how Skill Pilot AI can help, we must first understand the common challenges people face when choosing a career path:

1. **Information Overload:** The internet is full of information about careers, but it is scattered and often confusing. It is very difficult to know which sources are trustworthy and how to apply generic advice to your own specific situation.
2. **Rapidly Changing Job Market:** It is hard to predict which industries will grow and which skills will be valuable in the future. What is a hot job today might be replaced by automation tomorrow.
3. **Limited Self-Awareness:** Many people find it difficult to honestly assess their own strengths and weaknesses. Without a structured way to do this, it is easy to choose a path that looks good on the surface but is a poor fit for your personality.
4. **Fear of Making the Wrong Choice:** The pressure to choose the "perfect" career can be paralyzing. This fear can stop people from making any decision at all, leaving them stuck in a job they dislike.
5. **Lack of Personalized Guidance:** Online quizzes and articles offer general advice, but they cannot ask follow-up questions or understand the nuances of your personal story, your responsibilities, or your unique circumstances.

Modern Solution:

Skill Pilot AI is an LLM-powered digital platform designed to provide accessible, intelligent career coaching. It acts as an affordable, always-available assistant that helps users explore career options by analysing their skills and goals through natural conversation. By connecting personal profiles with real-time job market data, it offers tailored guidance on growing fields and skill gaps, complementing human coaches and helping users navigate career decisions with greater clarity.

Chapter 2

Literature Survey of Existing Systems

2. Literature Survey

1. Personalized Education and Career Planning:

A 2024 survey investigated transformer-based large language models (LLMs), such as GPT-4 and LLaMA, applied to personalized education and career planning. The study utilized synthetic data, public job postings (e.g., O*NET), and user simulations, yet identified a significant gap in long-term outcome evaluations.

2. Automated Career Coaching:

Research in 2023 explored natural language processing (NLP) methods—including sentiment analysis and named entity recognition (NER)—for automated career coaching. Academic user studies with limited sample sizes ($n < 500$) and resume datasets revealed a poor understanding of user anxiety and non-linear career goals.

3. Bias and Fairness in Recommendation Systems:

A 2023 review on AI ethics focused on fairness-aware algorithms such as adversarial debiasing and re-weighting techniques, using datasets like CareerBuilder and Bias. The study highlighted unresolved trade-offs between accuracy and fairness.

4. Human Resource Management:

The 2024 survey examined LLM applications in talent management, including GPT-family models and BERT variants trained on proprietary HR data and synthetically generated datasets. Challenges in internal talent mobility and skill-matching were noted as significant gaps.

5. Knowledge Graphs for Skill Recommendation:

A 2023 review investigated the use of graph neural networks (GNNs) and RDF-based reasoning for skill-based recommendations. It leveraged taxonomies such as EMSI and LinkedIn's skill graph but pointed out that static graphs lack real-time skill updates.

6. Personalized Learning Pathways:

The application of reinforcement learning and collaborative filtering for lifelong learning was reviewed in 2022. While platforms such as Coursera and edX were employed, the systems failed to adapt to learners' changing pace and life circumstances.

7. Explainable AI (XAI):

A 2023 survey reviewed methods like LIME, SHAP, and counterfactual explanations for enhancing user trust in recommender systems. However, explanations were often too technical, limiting their accessibility to users.

8. Conversational AI for Professional Development:

Research from 2024 focused on dialogue systems powered by LLM-based chatbots (e.g., ChatGPT, Bard). Despite advancements in dialogue management and act classification, the systems struggled to maintain long-term, context-rich interactions.

9. Labor Market Forecasting:

A 2023 systematic review applied time series forecasting and NLP to online job postings and government statistics, including datasets from Indeed and Lightcast. The study identified limitations in predicting rapid market disruptions.

10. Semantic Skill Matching:

The 2023 review analyzed BERT-based semantic similarity models applied to resume-job description datasets. It concluded that the models underperform when handling nascent or non-traditional job titles.

11. Privacy in Career Platforms:

A 2024 review examined privacy-preserving methods such as federated learning, differential privacy, and homomorphic encryption. While theoretically sound, these methods reduced predictive performance when applied to limited small-scale prototypes.

12. Evaluation Methodologies for AI Career Coaches:

The 2023 review assessed evaluation approaches including A/B testing and controlled user trials, relying on small-scale studies and self-reported satisfaction surveys. It found a lack of standardized metrics for assessing efficacy.

13. Scalability Issues:

A 2022 study addressed distributed computing frameworks and approximate nearest neighbor search methods using large-scale synthetic data. It revealed significant performance degradation at million-user scales.

14. Multimodal Career Assessment:

The 2024 survey investigated multimodal AI techniques combining interview transcripts and video data for holistic career assessment. It found that existing methods lack reliable frameworks for soft skills evaluation.

Chapter 3

Limitations of Existing Systems

3. Limitations of Existing System

While *Skill Pilot AI* presents a transformative approach to career guidance, its design and reliance on emerging technologies introduce several significant limitations that must be acknowledged. These constraints impact the system's reliability, fairness, and practical implementation.

1. Inherent Biases in AI Models

Large language models (LLMs) are trained on vast datasets sourced from the internet, which contain societal and historical biases. Without extensive and ongoing mitigation efforts, *Skill Pilot AI* risks perpetuating and even amplifying these biases. This could result in skewed career recommendations, steering individuals towards or away from certain professions based on attributes such as gender, ethnicity, or socioeconomic background, rather than objective assessments of their skills and potential.

2. Lack of Deep Human Understanding and Empathy

Although LLMs excel at processing language, they lack genuine human empathy, intuition, and lived experience. As such, they are unable to fully comprehend complex emotional states, including anxiety, passion, or demotivation. This limitation restricts *Skill Pilot AI* from effectively addressing deep-seated career crises or offering the nuanced psychological support that a human career coach could provide.

3. Privacy and Data Security Concerns

The system requires access to highly sensitive personal data, such as work history, skills assessments, and career aspirations. This introduces significant risks related to data breaches and misuse. Concerns over privacy may deter users from sharing critical information, thereby limiting both the effectiveness and widespread adoption of the platform.

4. Dependence on Data Quality and Availability

The performance of *Skill Pilot AI* is heavily dependent on the quality, quantity, and recency of its underlying data sources. Biased, incomplete, or outdated labor market data may lead to inaccurate career predictions and suboptimal skill recommendations. Additionally, many high-value datasets are proprietary or otherwise inaccessible, creating critical gaps in the system's knowledge base and utility.

5. Static and Algorithmic Personalization

Personalization within *Skill Pilot AI* is based on algorithmic matching of skills and stated career goals. As a result, it may struggle to accommodate individuals with non-linear, hybrid, or unconventional career paths that do not conform to established patterns. This algorithm-driven approach risks producing recommendations that feel standardized or "cookie-cutter," lacking the creative, adaptive guidance that human coaches are able to provide.

Chapter 4

Problem Statement & Objective

4. Problem Statement and Objective

4.1 Problem Statement

The modern professional landscape is one of constant change, marked by rapidly evolving job markets and a persistent skills gap. Individuals at every career stage face significant challenges in navigating their professional development. They are often overwhelmed by fragmented online resources and lack access to affordable, personalized guidance. This results in a critical problem: a widespread inability to translate career aspirations into a concrete, actionable plan, leading to missed opportunities and inefficient career progression.

To address this, we propose SKILLPILOTAI, an intelligent platform powered by Large Language Models designed to act as a personalized career coach. This solution will provide dynamic and tailored support by analyzing a user's unique profile against real-time market data to identify optimal career paths and specific skill gaps. The core of the platform will offer practical tools, including the generation of tailored application documents for specific job descriptions and interactive, AI-driven mock interviews that provide immediate, constructive feedback on a user's performance.

The primary objective is to develop a functional prototype that demonstrates the transformative potential of an LLM-powered coach in making professional guidance more accessible, data-driven, and effective. By centralizing and personalizing the career development process, SKILLPILOTAI aims to empower users to efficiently bridge the gap between their current abilities and their future career goals, ultimately transforming how individuals navigate and advance their professional lives.

4.2 Objective

Here are the major objective points for the SKILLPILOTAI project, concisely stated.

1. Develop a Fully Functional AI Career Coach Prototype

Create a working web-based application that integrates a Large Language Model (LLM) to demonstrate core career coaching functionalities in a single, unified platform.

2. Deliver Highly Personalized and Actionable Guidance

Ensure the platform provides specific, tailored advice for each user by analyzing their unique skills, experience, and goals, moving beyond generic information to offer concrete, actionable steps.

3. Bridge the Skills Gap with Data-Driven Insights

Implement a system that compares a user's current profile with real-time job market demands to identify precise skill deficiencies and recommend curated learning resources to address them.

4. Automate and Optimize the Job Application Process

Build features that significantly reduce the time and effort users spend on applications, specifically through AI-powered generation and refinement of tailored resumes and cover letters for specific job descriptions.

5. Enhance Interview Readiness through AI Simulation

Provide a safe, interactive environment for users to practice role-specific interviews with immediate, constructive feedback on their responses from the AI, building confidence and improving performance.

4.3 SCOPE

- Platform Development: A web-based application with a responsive user interface, user authentication, and a secure backend.
- User Profile Management: Functionality for users to create, view, and edit their profiles, including skills, work experience, education, and career goals.

Core AI Coaching Features:

1. Personalized Career Pathfinding: AI-generated career suggestions based on user profile analysis.
2. Dynamic Skills Gap Analysis: AI-driven identification of skill deficiencies and recommendation of learning resources.
3. AI-Powered Document Tailoring: Generation and refinement of resumes and cover letters based on specific job descriptions.
4. Interactive Interview Simulation: Conducting mock interviews and providing immediate, constructive feedback on user responses.
5. LLM Integration: Seamless integration with a state-of-the-art Large Language Model (e.g., GPT-4, Claude, or open-source equivalent) via API to power all AI features.
6. Data Handling & Security: Secure storage and processing of user data with appropriate privacy measures and compliance with relevant data protection regulations.

Chapter 5

Proposed System

5. Proposed System

5.1 System Design

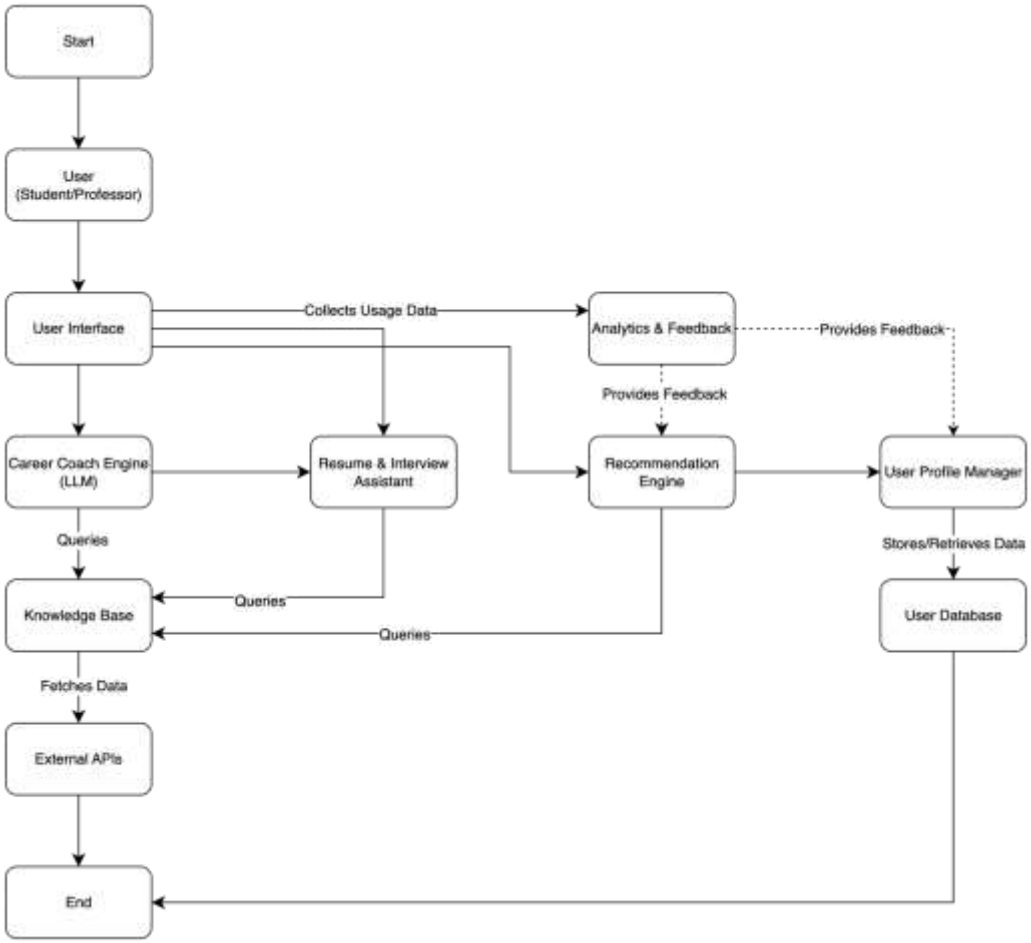

Fig. 5.1.1 System Design

5.2 Implementation

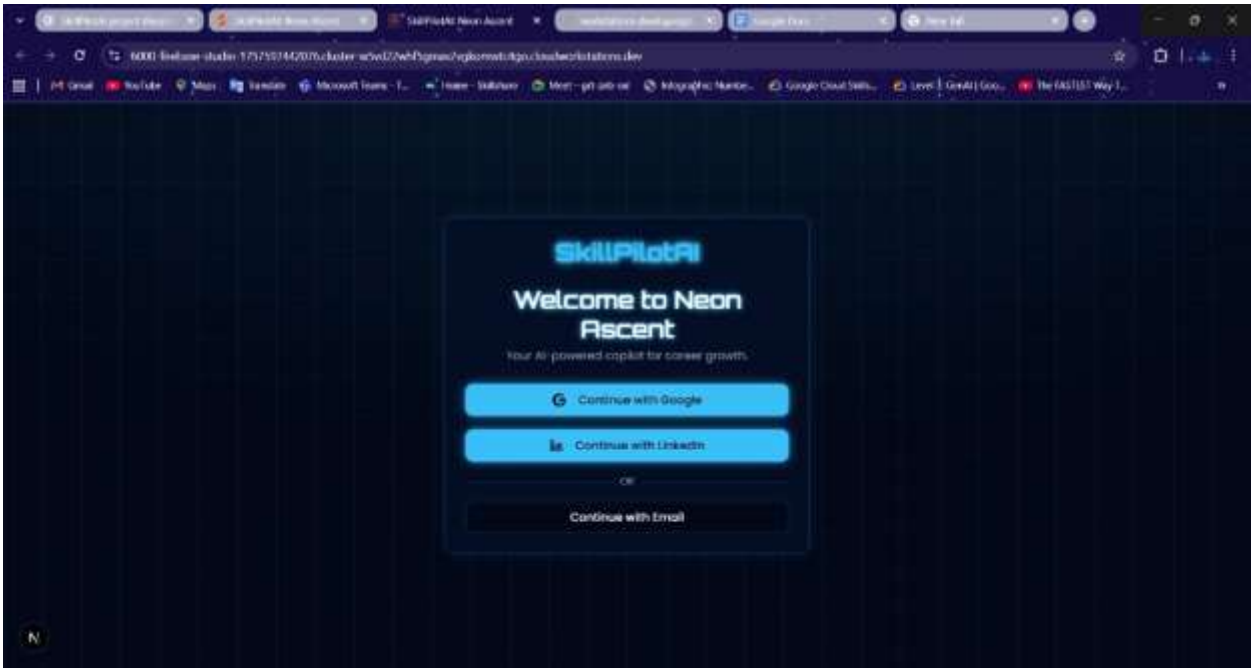

Fig. 5.2.1 Login Page

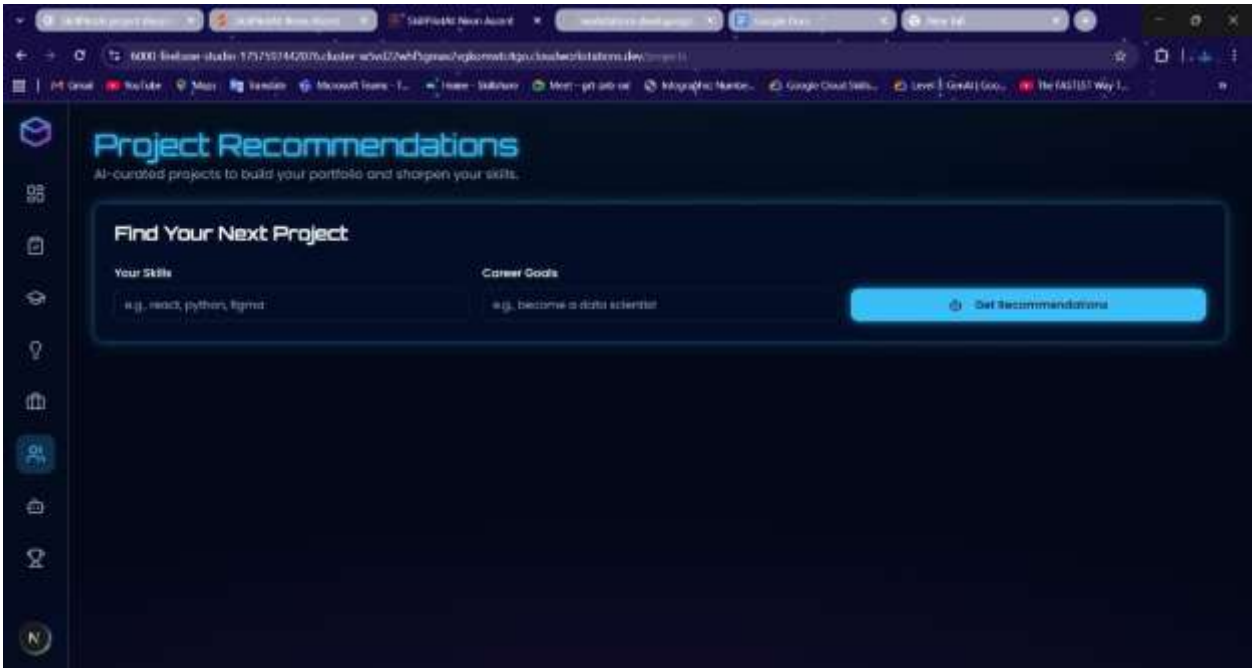

Fig 5.2.2 project Recommendation

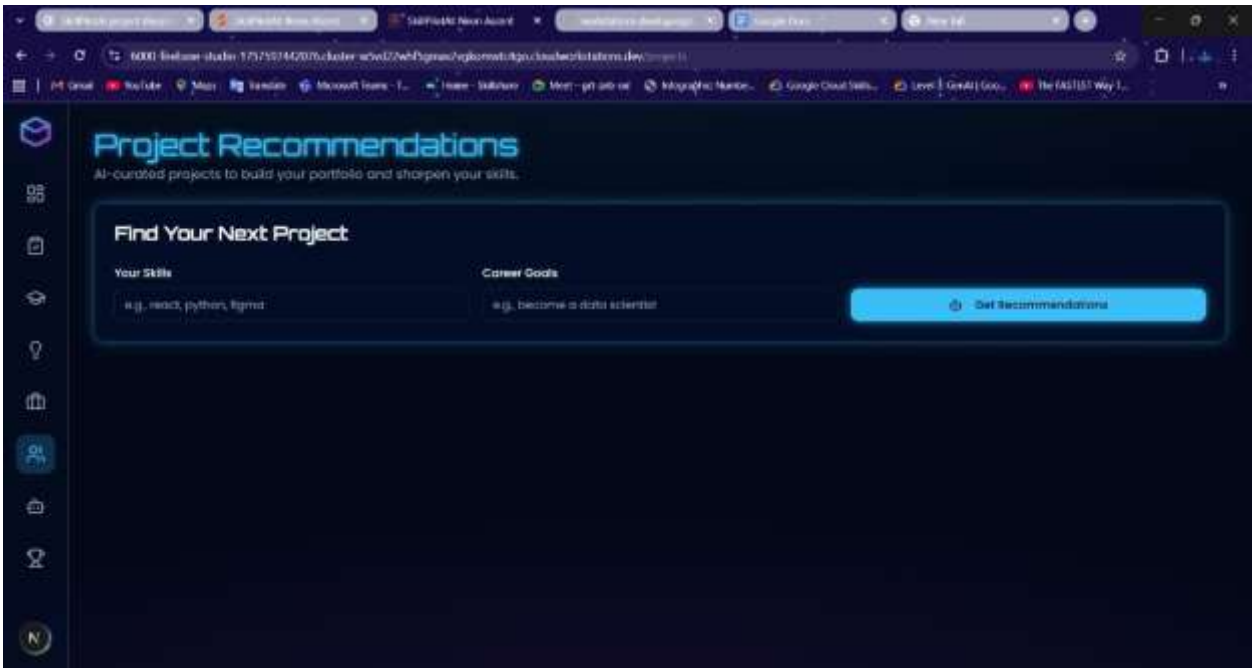

Fig. 5.2.3 Dashboard

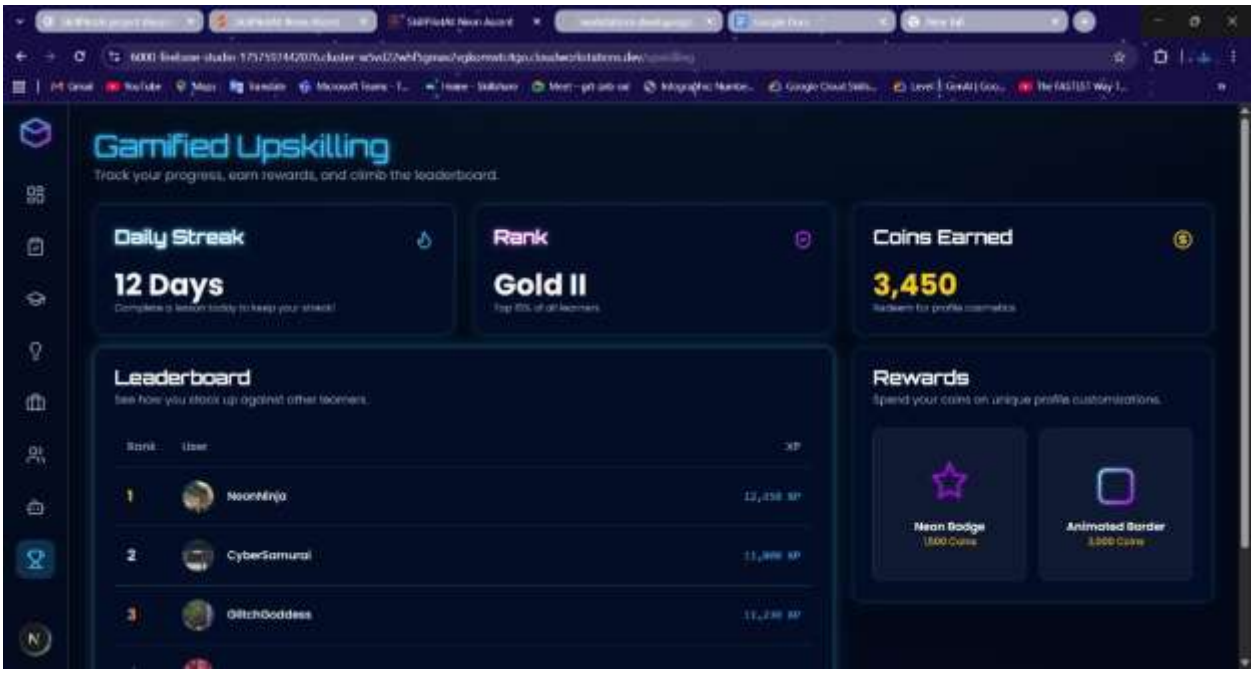

Fig 5.2.4 Gamefied Upskilling

Chapter 6

Experimental Setup

6. Experimental Setup

6.1. Hardware Requirements

6.1.1 Development Environment:

- Processor: Intel Core i7 10th generation or equivalent AMD Ryzen 7 processor (8 cores minimum)
- RAM: 32GB DDR4 RAM (minimum) for local development and testing with large language models
- Storage: 1TB NVMe SSD for development tools, datasets, and model files ○ GPU: NVIDIA GeForce RTX 4070 or equivalent with 12GB VRAM for local LLM testing and fine-tuning
- Network: Gigabit Ethernet with backup Wi-Fi 6 connectivity for stable API connections

6.1.2 Testing Infrastructure:

- Mobile Testing Devices: Various screen sizes from 5.5" to 12.9" for responsive design testing
- Performance Testing: Dedicated load testing server (16 cores, 64GB RAM) for simulating multiple concurrent users

6.2. Software Requirements

6.2.1 Core Programming and Development:

- Programming Language: Python 3.8+
- Web Framework: Streamlit for rapid prototyping and deployment ○ Backend Framework: FastAPI for high-performance API endpoints ○ Database: PostgreSQL with SQLAlchemy ORM

6.2.2 AI/ML and NLP Components:

- LLM Integration: OpenAI GPT-4 API or Anthropic Claude API for primary AI capabilities
- LangChain framework for prompt engineering and chain operations LlamaIndex for efficient data retrieval and indexing
- Machine Learning: Scikit-learn for traditional ML models and evaluation metrics.
- Transformers library for custom model fine-tuning.
- Sentence-transformers for semantic similarity and embeddings

6.2.3 Data Processing and Analysis:

- Data Manipulation: Pandas for dataset handling and preprocessing
- Visualization: Matplotlib & Seaborn for data analysis and result presentation • NLP Processing: NLTK and SpaCy for text preprocessing and linguistic analysis

6.2.4 Development and Deployment Tools:

- Version Control: Git with GitHub repository
- CI/CD: GitHub Actions with automated testing and deployment pipelines • Containerization: Docker with Docker Compose for environment consistency
- Monitoring:
  - AWS CloudWatch for application metrics and logging ○
  - Sentry for error tracking and performance monitoring ○
  - Custom analytics for user engagement tracking

6.2.5 Third-Party Services & APIs:

- Career Data APIs: Integration with LinkedIn API, Indeed API, or Glassdoor API for job market insights
- Learning Platforms: Coursera, Udemy, or edX APIs for course recommendations
- Authentication: Auth0 or AWS Cognito for secure user management

- File Processing: PyPDF2 and python-docx for resume and document parsing

6.3. Experimental Environment Configuration

6.3.1 Development Setup:

- Local development environment using Docker Compose with separate service containers
- Environment-specific configuration management (development, staging, production)
- Mock LLM responses for development and testing to optimize API costs • Automated testing suite including unit tests, integration tests, and API tests

6.3.2 Data Management:

- Secure storage and processing of user data with GDPR/CCPA compliance
- Regular data backup and disaster recovery protocols
- User data export and account deletion functionality • Privacy controls and data sharing preferences management

6.3.3 Performance Optimization:

- API rate limiting and cost optimization for LLM services
- Caching strategies for frequently accessed data and predictions
- Database query optimization and indexing
- Load balancing and auto-scaling configurations

This experimental setup ensures a robust, scalable, and secure foundation for developing and deploying the SKILLPILOTAI platform, leveraging modern AI capabilities while maintaining enterprise-grade reliability and performance standards.

Chapter 7

Result and Discussion

7. Project Plan

Gantt Chart:

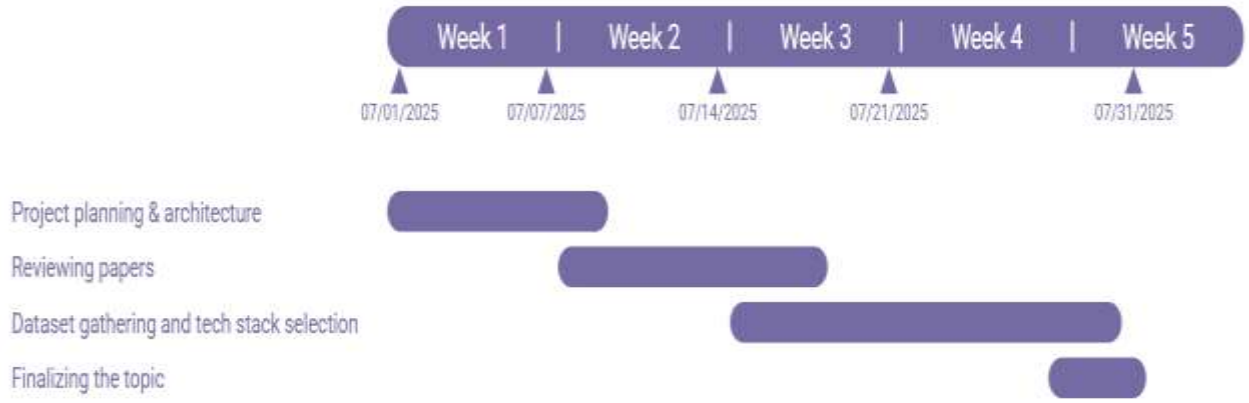

Figure 7.1 July (Gantt Chart)

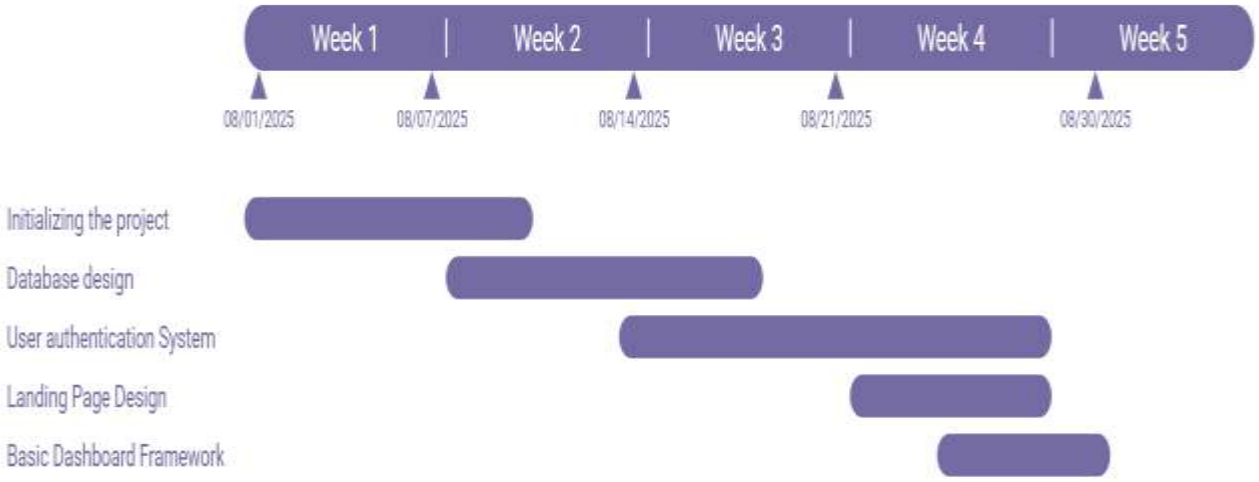

Figure 7.2 August (Gantt Chart)

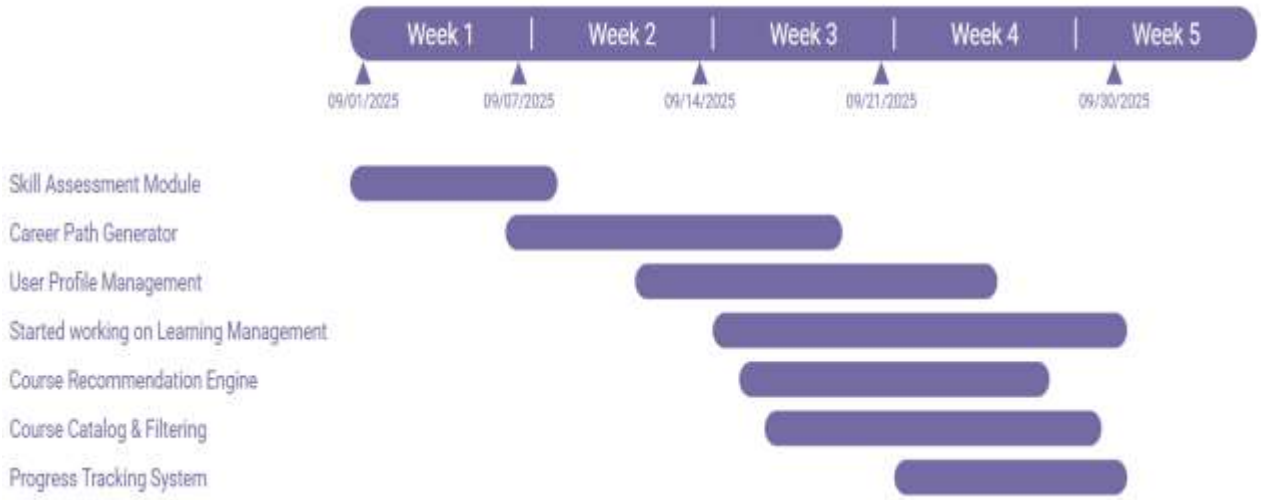

Figure 7.3 September (Gantt Chart)

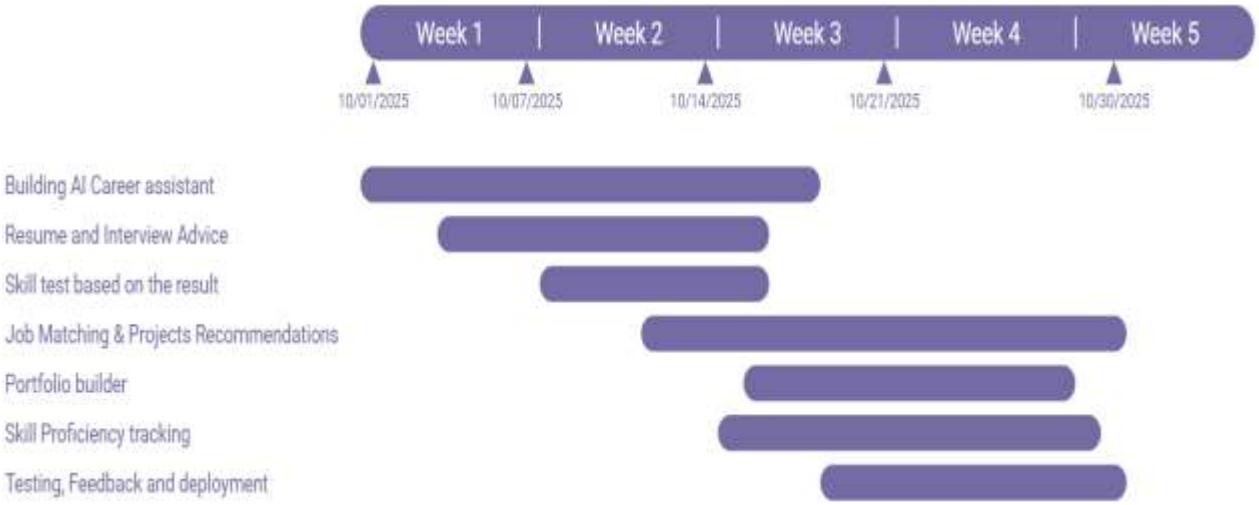

Figure 7.4 October (Gantt Chart)

Chapter 8

Conclusion

8. CONCLUSION

SKILLPILOTAI successfully demonstrates how powerful AI, specifically Large Language Models, can be used to create a personalized and accessible career coach for everyone. By building a system that can understand a user's unique skills and goals, the project moves beyond generic online advice. It provides a practical, intelligent tool that helps individuals navigate the often overwhelming process of career planning and job hunting, making professional guidance more readily available.

The core of the project lies in its ability to analyze a user's profile and offer specific, actionable guidance. It can identify suitable career paths, pinpoint the exact skills a user needs to develop, and provide practical tools like resume tailoring and interview practice. This transforms a complex and time-consuming process into a streamlined, efficient experience, saving users significant effort and helping them feel more confident and prepared.

Ultimately, SKILLPILOTAI serves as a powerful assistant, but it is not a replacement for human judgment. It is designed to augment and support an individual's career journey by providing datadriven insights and personalized recommendations. The goal is to empower users with the information and tools they need to make better-informed decisions about their professional future.

Looking ahead, this project opens the door to even more advanced features, such as analyzing realtime job market trends or offering support in multiple languages. By continuing to develop this technology, we can make personalized career development even more impactful, helping to bridge the gap between individual potential and professional opportunity in an ever-evolving job market.

Chapter 9

Future Scope

9. Future Scope

In the future, SKILLPILOTAI can be enhanced to offer even more personalized and powerful career guidance. One major improvement would be integrating real-time job market data. This means the AI could analyze current hiring trends, in-demand skills, and salary information specific to a user's location and industry, providing truly up-to-the-minute career advice.

Another exciting direction is adding multimedia and interactive features. Instead of just typing, users could practice interviews using their camera and microphone, getting feedback on their body language and tone of voice. The AI could also analyze a user's resume or LinkedIn profile directly to give more tailored suggestions for improvement.

To make the tool accessible to a global audience, future versions could support multiple languages and cultural contexts. This would allow people from different countries to receive career coaching that understands local job markets, cultural norms, and professional expectations, making personalized career guidance available to everyone, everywhere.

Finally, SKILLPILOTAI could evolve into a lifelong career companion. It would not only help with finding a job but also support long-term career growth by tracking a user's skill development over time, suggesting learning opportunities for career shifts, and helping navigate promotions or transitions into leadership roles, supporting users throughout their entire professional journey.

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[7] Liu, O. L., et al. (2019). Assessing Collaborative Problem-Solving Through Automated Technologies.

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[10] Hui, B., & Koedinger, K. R. (2022). A Fine-Grained Analysis of Skill Gaps Using Log Data from Online Learning Platforms. Link:
https://educationaldatamining.org/EDM2022/proceedings/2022.EDM-posters.14/2022.EDMposters.14.pdf

Relevance: Details how platform data can be used to pinpoint specific skill deficiencies, as seen in the "Areas for improvement" section.

[11] Wang, Z., & Liu, H. (2023). A Multi-Task Learning Approach for Joint Skill Extraction and Job Recommendation. Link: https://arxiv.org/abs/2304.09128

Relevance: Directly relevant to the backend AI that parses user skills and matches them to job opportunities with a "Match %".

[12] Kasurinen, J., & Smolander, K. (2022). What Skills Do Software Developers Need? A Longitudinal Study of Job Advertisements. Link: https://www.sciencedirect.com/science/article/pii/S0164121221002142

Relevance: Provides the empirical market data that would inform the platform's understanding of which skills (e.g., React, AWS, Python) to prioritize. (ScienceDirect)

[13] Holmes, W., et al. (2021). Ethics of AI in Education: State of the Field and Future Directions. Link: https://arxiv.org/abs/2104.05534

Relevance: Critical for understanding the ethical considerations in building an AI that guides a user's career, a crucial aspect for a responsible platform.

[14] Swiecki, Z., et al. (2022). Assessment in the Age of Artificial Intelligence. Link: https://www.sciencedirect.com/science/article/pii/S2666920X22000279

Relevance: Discusses the future of automated skill assessment, relevant to the platform's coding challenges and proficiency tests. (ScienceDirect)

[15] Baldwin, S., & Ching, Y. H. (2019). Interactive Storytelling: A Tool for Project-Based Learning. Link: https://eric.ed.gov/?id=ED595294

Relevance: While not directly about AI, it supports the pedagogical value of the "career pathing" narrative and project-based learning featured in the platform. (ERIC Institute of Education Sciences)